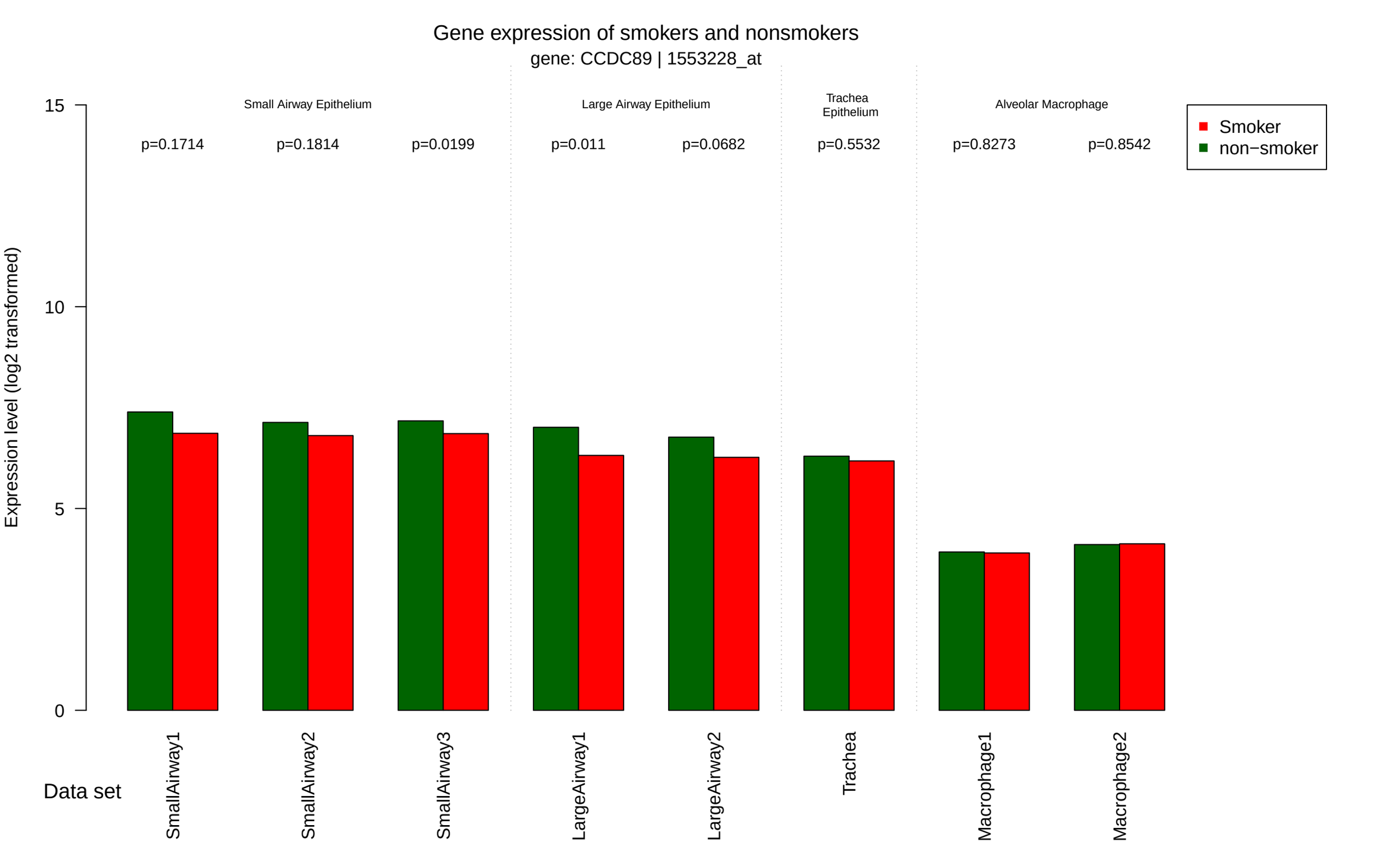
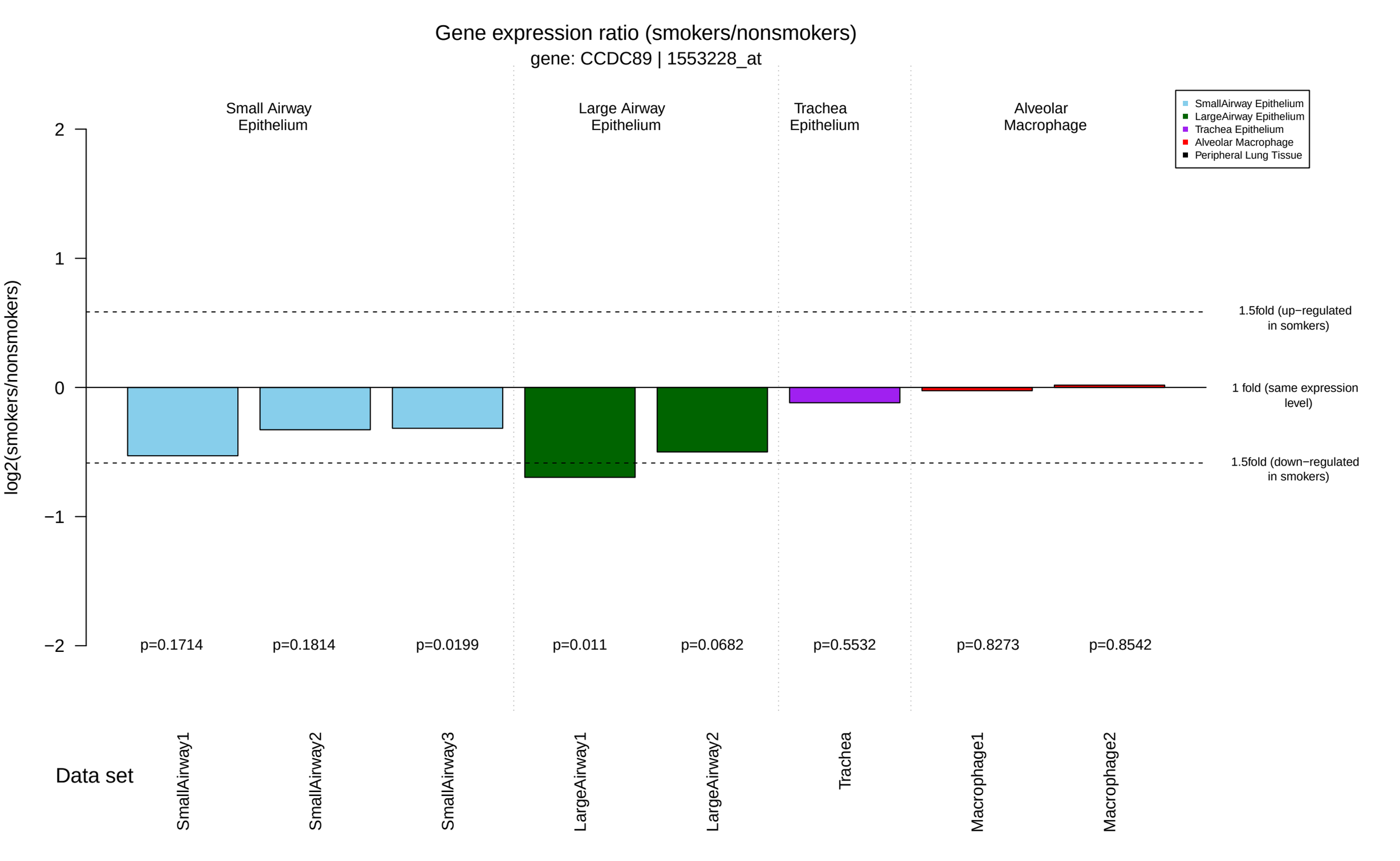
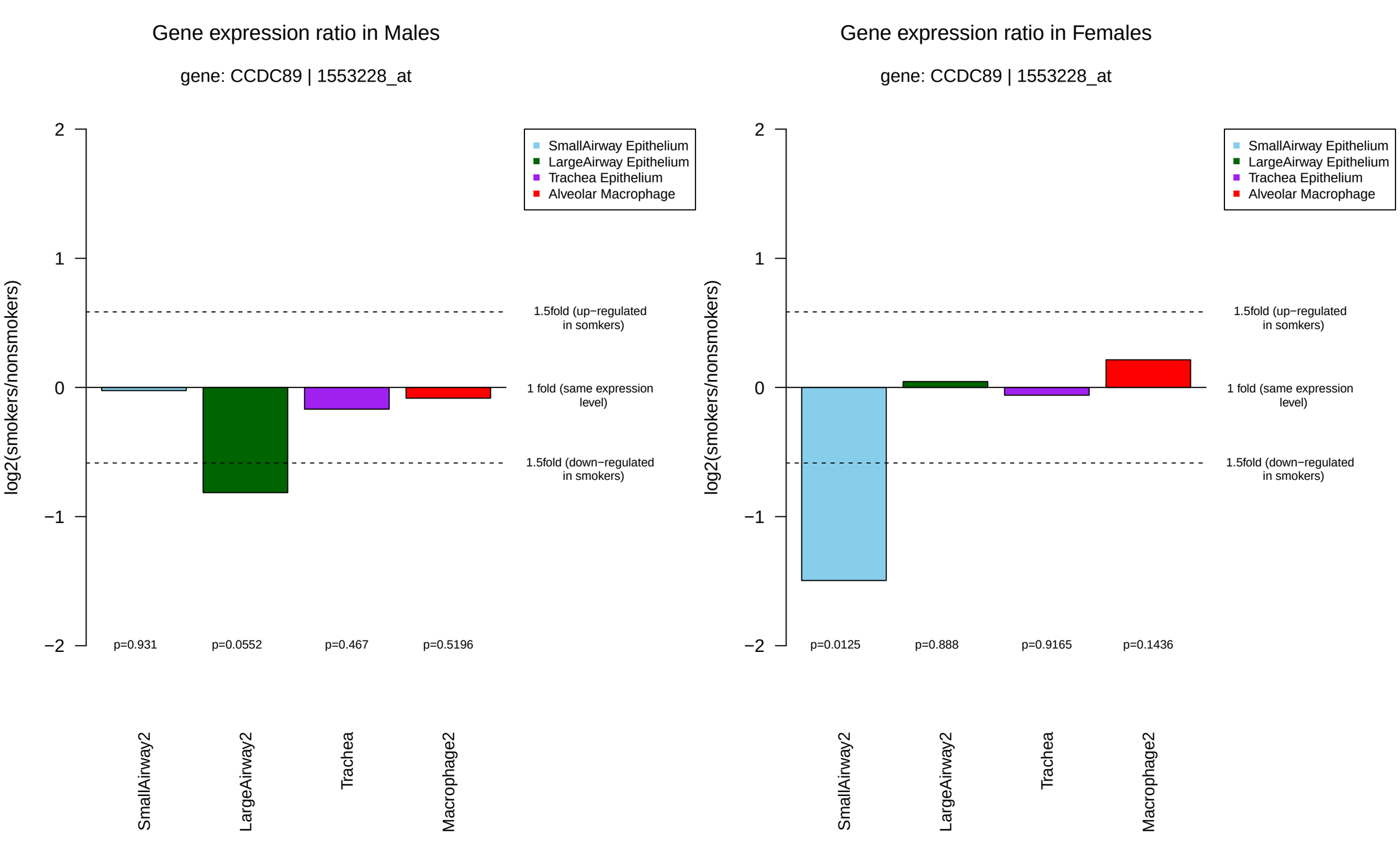
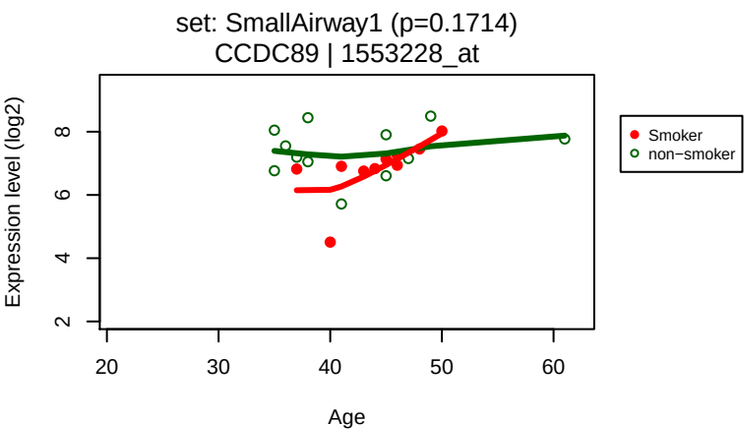
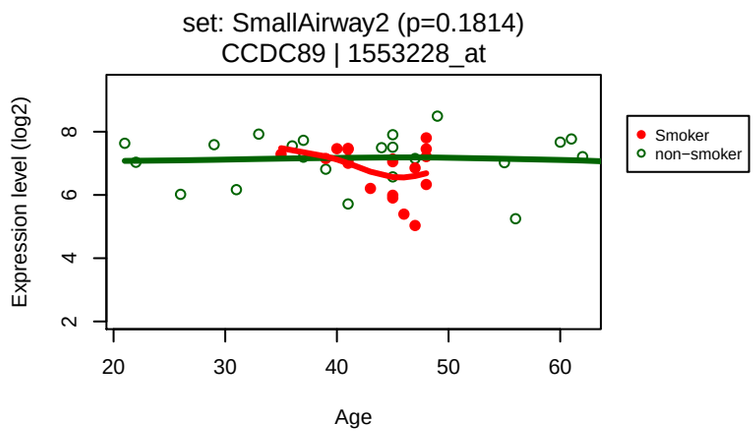
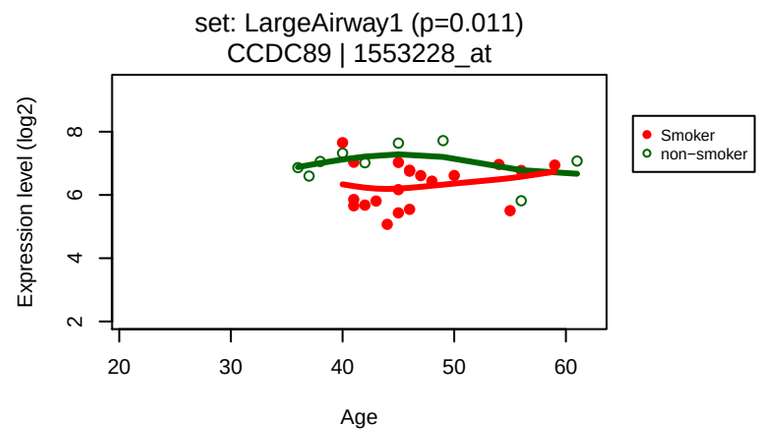

Gene expression of smokers and nonsmokers
gene: CCDC89 | 1553228_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1714
p=0.1814
p=0.0199
p=0.011
p=0.0682
p=0.5532
p=0.8273
p=0.8542
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

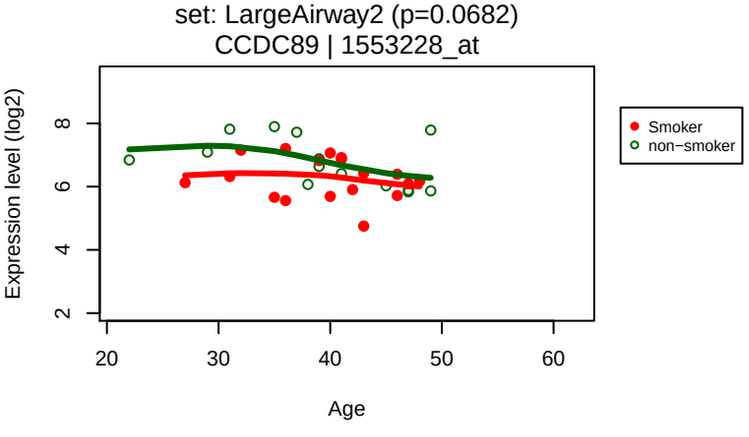

Gene expression ratio (smokers/nonsmokers)
gene: CCDC89 | 1553228_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1714
p=0.1814
p=0.0199
p=0.011
p=0.0682
p=0.5532
p=0.8273
p=0.8542
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

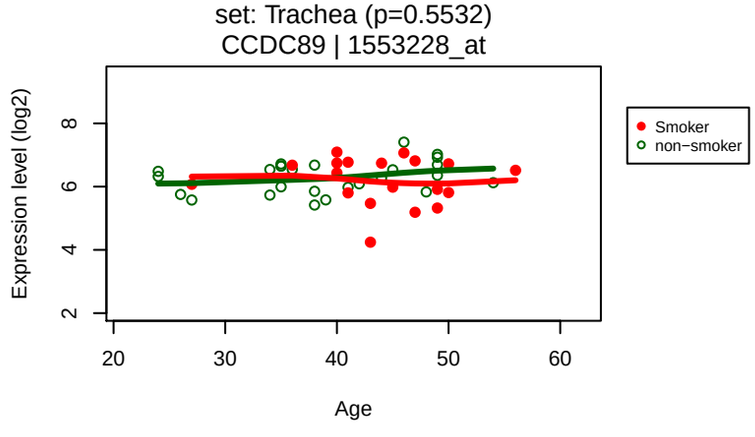

Gene expression ratio in Males
gene: CCDC89 | 1553228_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.931
p=0.0552
p=0.467
p=0.5196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CCDC89 | 1553228_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0125
p=0.888
p=0.9165
p=0.1436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

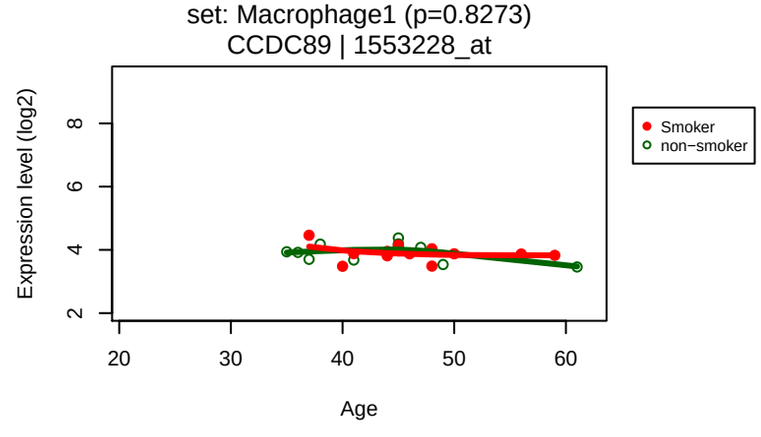

set: SmallAirway1 (p=0.1714)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker

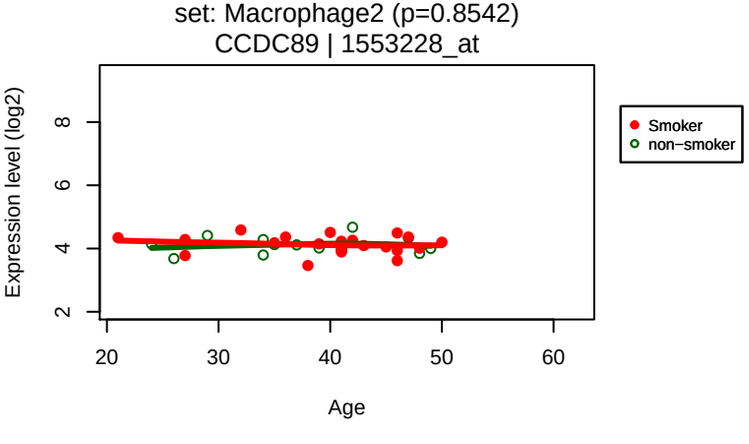

set: SmallAirway2 (p=0.1814)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker

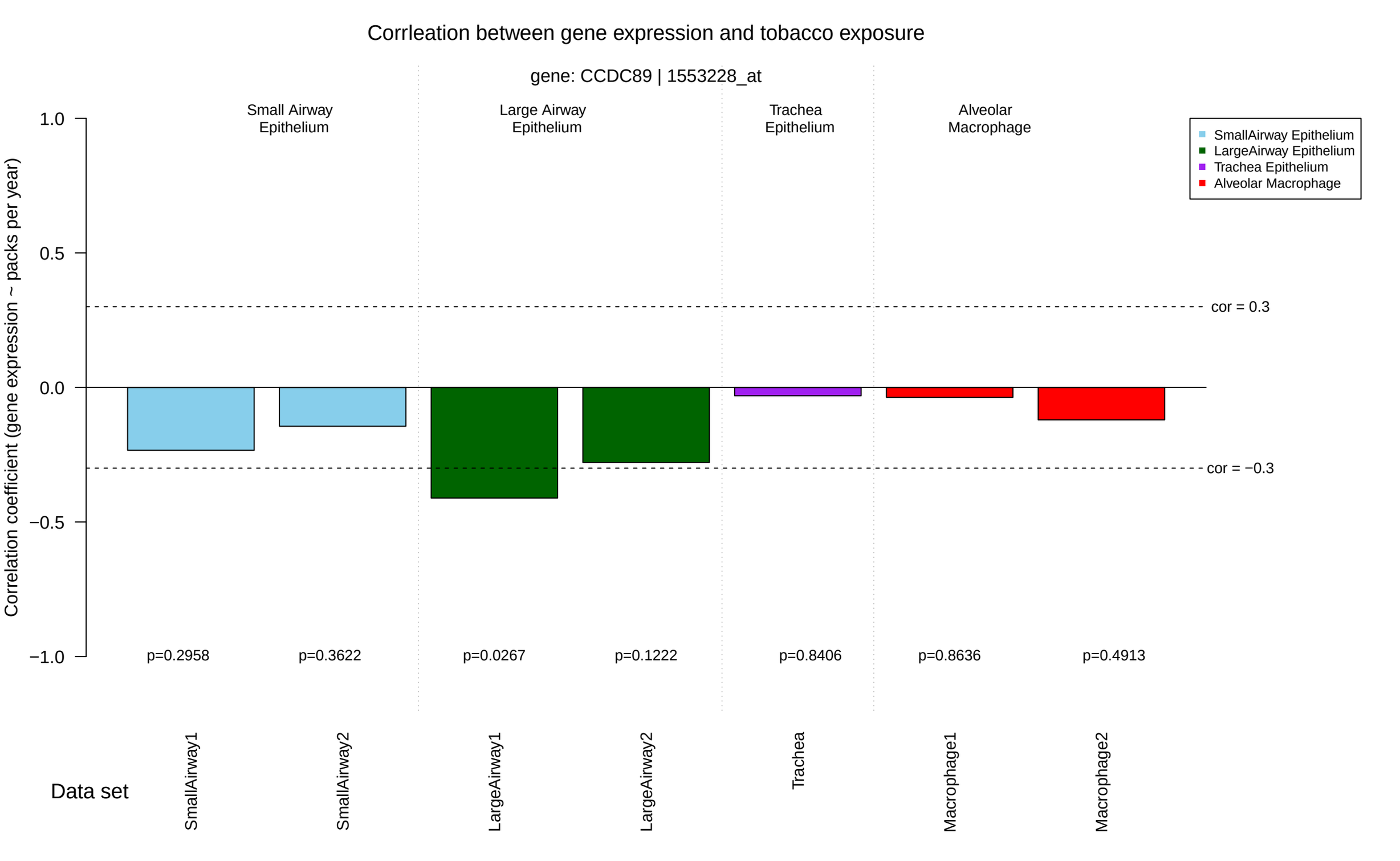

set: LargeAirway1 (p=0.011)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0682)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5532)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8273)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8542)
CCDC89 | 1553228_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CCDC89 | 1553228_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2958
p=0.3622
p=0.0267
p=0.1222
p=0.8406
p=0.8636
p=0.4913
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2